Between Algorithms and Knowledge: Artificial Intelligence in Teacher Education in Higher Education

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Abstract

The advancement of artificial intelligence (AI) technologies, particularly generative tools such as ChatGPT, has profoundly impacted educational processes. In the context of teacher education, these impacts go beyond the technical use of AI and raise discussions around epistemology, ethics, and pedagogical culture. To analyze the scientific literature on the use of artificial intelligence in the education of higher education teachers, with an emphasis on the pedagogical, ethical, and cultural implications associated with its integration. This is an Integrative Literature Review, developed based on the PICo strategy: P (higher education teachers), I (use of artificial intelligence in teacher education), Co (educational contexts). The search was conducted in the SciELO, LILACS, PubMed, Web of Science, and Scopus databases, covering the period from 2019 to 2025. After applying the eligibility criteria and analyzing the data using the PRISMA model, 12 articles were included in the final corpus and systematized in a summary table, with evidence levels classified according to the Joanna Briggs Institute. The selected studies highlight the growing interest in AI in teacher education over the past three years, with a predominance of qualitative approaches and publications from Latin America, Western Europe, and North America. The literature emphasizes both innovative educational experiences and concerns about the instrumentalization of teaching, the superficiality of pedagogical mediation, and the ethical risks of uncritical AI usage. The analysis reveals a tension between two approaches: one that promotes AI as an automation tool, and another that advocates for its critical and ethical mediation, valuing the teacher's role as author and educator. These findings underscore the urgency of policies and continuing education programs that incorporate AI with pedagogical intentionality. The presence of AI in teacher education demands not only technical skills but also ethical discernment, critical reflection, and commitment to emancipatory educational practices. This study contributes to expanding the debate and supporting professional development strategies aligned with the complexities of contemporary teaching.